SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CSA1717 – ARTIFICIAL INTELLIGENCE

COMMON COURSE ASSIGNMENT

Intelligent Traffic Congestion Prediction and Smart Route Recommendation

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Particular	**Details**
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Programme	B.Tech
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Faculty Name	Dr. Anoop M
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Bloom's Taxonomy Level	L4 – Analyze, L5 – Evaluate, L6 – Create
SDG Mapping	SDG 9 (Industry, Innovation, and Infrastructure); SDG 11 (Sustainable Cities and Communities)
Industry / Societal Relevance	Smart mobility, traffic management and sustainable transport

1. Problem Statement

The objective of this assignment is to develop an AI-based Intelligent Transportation System capable of predicting traffic congestion and recommending an efficient route between a source and destination.

The proposed system performs the following:

Collect and preprocess road-network and traffic-related data.

Calculate traffic-related indicators from road capacity and average traffic.

Use an SVM classification model to predict whether a road segment is congested.

Represent the road network as a weighted graph.

Use Dijkstra's algorithm to determine an efficient route between a source and destination.

Integrate SVM congestion predictions into the route cost.

Evaluate the prediction and route recommendation performance.

Consider efficiency, safety, sustainability, environmental impact and responsible use of transportation data.

The supplied dataset contains road characteristics including road type, length, number of lanes, speed limit, road capacity, bus/bicycle lanes, surface condition, geographical coordinates, district, toll rate and average daily traffic.

C. PROBLEM STATEMENT

Problem / Case / Design Challenge

Urban traffic congestion increases travel time, fuel consumption and vehicle emissions. Selecting a suitable route becomes difficult when traffic conditions vary across different road segments.

The objective of this assignment is to design and develop an AI-based intelligent transportation system that predicts traffic congestion and recommends an efficient route between a specified source and destination.

The proposed system uses the supplied road-network dataset containing 200 road segments with traffic, capacity, road characteristics and geographical information.

Since the dataset does not provide an explicit congestion class, a Volume-to-Capacity (V/C) ratio is calculated:

```
V/C=Average Daily TrafficRoad Capacity
```

For this assignment:

```
V/C>1
```

is considered congested, while

```
V/C≤1
```

is considered non-congested.

Using this derived classification, an SVM model is trained to identify congestion conditions from available road and traffic attributes.

The predicted congestion information is then integrated with a graph-based route-selection system. Each road segment is represented as an edge, and Dijkstra's shortest-path algorithm is used to identify a route with lower estimated travel cost.

The final system therefore combines:

```
Road Dataset → Preprocessing → Congestion Feature Engineering → SVM Prediction →
Graph Construction → Dijkstra → Smart Route
```

2. Problem Understanding and Formulation

2.1 What is the problem?

The problem is to predict whether individual road segments are experiencing congestion and then use those predictions to recommend an efficient route.

Traditional shortest-path routing may select a geographically short road without considering congestion. Therefore, the proposed system combines machine learning with graph search.

The system has two main intelligent components:

Component 1 – SVM Congestion Prediction

SVM analyzes road and traffic characteristics and predicts the congestion condition.

Component 2 – Dijkstra Smart Routing

Dijkstra evaluates the road network using congestion-aware edge costs and identifies a low-cost path.

2.2 Expected Outcomes

The expected outputs are:

Congestion classification for road segments.

SVM prediction accuracy and performance metrics.

Congestion risk information for each road.

Weighted road graph.

Recommended source-to-destination route.

Estimated route distance and travel time.

Comparison between conventional and congestion-aware routing.

2.3 Available Information

The dataset provides 200 road records with 18 attributes. It includes traffic volume, road capacity and geographical coordinates, which are particularly important for this project.

For example, the dataset contains roads such as RD-0001, RD-0002 and RD-0003 with their respective lengths, capacities, speed limits and traffic values.

2.4 Major Assumptions

A road is considered congested when its traffic demand exceeds its hourly capacity:

```
V/C>1
```

Road segments can be represented as graph edges.

Geographical endpoints within a predefined distance can be treated as connected intersections.

The speed limit is used as the free-flow speed estimate.

SVM predictions are used to estimate congestion risk.

Dijkstra minimizes the resulting route cost.

The dataset represents a simplified road network and is used for academic demonstration.

Congestion Distribution

Congestion Class	Number of Roads	Percentage
Non-congested	55	27.5%
Congested	145	72.5%
Total	200	100%

3. Application of Course Knowledge

2.1 Artificial Intelligence

Artificial Intelligence is applied to automate congestion classification and route selection.

The proposed system combines:

Machine learning

Classification

Feature preprocessing

Graph representation

Shortest-path search

Decision-making

2.2 Support Vector Machine

SVM is a supervised machine-learning algorithm that finds a decision boundary separating different classes.

For a linear classifier:

$$fx=wTx+b$$

The objective is to find a suitable hyperplane that maximizes the separation between classes.

For nonlinear road-congestion patterns, an RBF kernel is used:

$$Kxixj=e-\gamma\|xi-xj\|2$$

The SVM predicts whether a road segment is:

0 – Non-congested

1 – Congested

2.3 Congestion Feature Engineering

The most important derived parameter is:

$$V/C=\text{Average Daily TrafficCapacity}$$

For example, RD-0001 has average daily traffic of 10,833 and capacity of 5,384 vehicles/hour.

Therefore:

$$V/C=108335384\ ;\ V/C\approx2.01$$

Since:

$$2.01>1$$

the road is classified as congested according to the project's rule.

2.4 Congestion Distribution

Applying the above rule to the complete dataset gives:

This shows that the supplied dataset contains more congested than non-congested road segments.

4. Solution / Design / Methodology

3.1 Proposed System Architecture

The proposed architecture is:

Road Dataset

↓

Data Preprocessing

↓

Feature Engineering

↓

V/C Ratio Calculation

↓

Congestion Label Generation

↓

SVM Training

↓

Congestion Prediction

↓

Road Graph Construction

↓

Congestion-Aware Edge Weight

↓

Dijkstra Algorithm

↓

Optimal Route

↓

Route Evaluation

3.2 Module 1 – Data Preprocessing and Congestion Analysis

The first module loads the CSV dataset.

Steps

Read road_network.csv.

Check missing values.

Convert maintenance dates.

Encode categorical variables.

Calculate V/C ratio.

Generate congestion labels.

Separate features and target.

Scale numerical features.

The supplied dataset contains no missing values across the 18 fields in the inspected CSV.

3.3 Module 2 – SVM Congestion Prediction

The second module applies SVM.

Input Features

The model can use:

Length

Number of lanes

Speed limit

Capacity

Average traffic

Bus-lane availability

Bicycle-lane availability

Toll rate

Road type

Surface condition

District

The categorical variables are converted using one-hot encoding, while numerical variables are standardized.

Training

The dataset is divided into:

80% training data

20% testing data

The SVM uses an RBF kernel.

SVM Prediction Results

Metric	Result
Accuracy	72.5%
Precision	72.5%
Recall	100%
F1-score	84.1%

3.4 SVM Algorithm

Algorithm

Load processed dataset.

Separate features and target.

Encode categorical features.

Standardize numerical features.

Split data into training and testing sets.

Train SVM classifier.

Predict congestion on test data.

Calculate accuracy, precision, recall and F1-score.

Train the final model using the available dataset.

Generate congestion predictions for route planning.

3.5 SVM Results

Using the supplied dataset and the above preprocessing configuration, the preliminary 80:20 evaluation produced:

Five-fold stratified cross-validation produced an average accuracy of approximately:

```
73%
```

Interpretation

The SVM successfully identifies congested roads with high recall. However, the relatively moderate accuracy indicates that the available road attributes are not sufficient for highly reliable congestion prediction.

This is expected because the dataset does not contain real-time traffic measurements or an independently observed congestion label.

3.6 Module 3 – Smart Route Recommendation Using Dijkstra

After predicting congestion, the road network is converted into a graph.

Graph Representation

Node: road endpoint/intersection

Edge: road segment

Edge weight: estimated travel cost

The dataset provides starting and ending latitude/longitude for every road segment.

Because exact endpoint coordinates are not repeated as explicit intersections, nearby endpoints are connected using a geographical proximity assumption.

3.7 Dijkstra's Algorithm

Dijkstra's algorithm finds the minimum-cost path from a source node to a destination node in a weighted graph with non-negative edge weights.

The basic relaxation operation is:

```
dv=min■dvdu+wuv
```

where:

```
= current distance to node dv ; v
= distance to node du ; u
= cost of travelling from to wuv ; u ; v
```

3.8 Congestion-Aware Edge Weight

Instead of simply using road distance, the system uses estimated travel time.

Free-flow travel time is:

```
Te=LeVe×60
```

where:

```
= road length in km Le
= speed limit in km/h Ve
= travel time in minutes Te
```

The SVM decision value is converted into a monotonic congestion-risk score between 0 and 1.

The congestion-aware cost is then:

```
Ce=Te1+0.6Re

where is the SVM-derived congestion-risk score. Re
```

Thus, a road predicted to have higher congestion receives a larger routing cost.

3.9 Dijkstra Route Recommendation Algorithm

Algorithm

Load road network.

Create graph nodes from road endpoints.

Create edges representing road segments.

Calculate free-flow travel time.

Obtain SVM congestion prediction/risk.

Modify edge cost according to congestion risk.

Select source node.

Select destination node.

Initialize source distance as zero.

Select the unvisited node with minimum distance.

Relax neighboring nodes.

Repeat until destination is reached.

Trace predecessor nodes.

Return recommended route.

5. Use of Modern Tools and Implementation Outputs

The following tools can be used for implementation:

Tool	Purpose
Python	Main implementation
Pandas	Dataset handling
NumPy	Numerical computation
Scikit-learn	SVM implementation
NetworkX	Graph and Dijkstra implementation
Matplotlib	Graphs and visualization
Visual Studio Code (VS Code)	Implementation and Testing
Git/GitHub	Version control

The following output screenshots are retained from the submitted assessment to document the actual implementation and execution evidence.

```
[5 rows x 18 columns]

Dataset Shape: (200, 18)
PS C:\Users\yanar\Documents\Traffic_Congestion_Project> py traffic_prediction.py
   road_id      road_name      road_type  length_km  lanes  ...  latitude_end  longitude_end    district  toll_rate_rp avg_daily_traffic
0  RD-0001  Jl. Segment 1  Jalan Kolektor        4.3      4  ...     -6.187271     106.937674         PIK             0             10833
1  RD-0002  Jl. Segment 2    Jalan Arteri        4.3      4  ...     -6.246220     106.771962       Depok             0              6921
2  RD-0003  Jl. Segment 3     Jalan Lokal        1.8      2  ...     -6.264098     106.831007   Tangerang             0              1085
3  RD-0004  Jl. Segment 4  Jalan Kolektor        1.8      4  ...     -6.122268     106.958261  Mangga Dua             0              6145
4  RD-0005  Jl. Segment 5       Jalan Tol       22.7      4  ...     -6.341074     106.856648    Sudirman          7000              8023

[5 rows x 18 columns]

Dataset Shape: (200, 18)
```

Figure 1. Dataset loading output.

```
Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 18 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   road_id                     200 non-null    object
 1   road_name                   200 non-null    object
 2   road_type                   200 non-null    object
 3   length_km                   200 non-null    float64
 4   lanes                       200 non-null    int64
 5   speed_limit_kmh             200 non-null    int64
 6   capacity_vehicles_per_hour  200 non-null    int64
 7   has_bus_lane                200 non-null    int64
 8   has_bike_lane               200 non-null    int64
 9   surface_condition           200 non-null    object
 10  last_maintenance_date       200 non-null    object
 11  latitude_start              200 non-null    float64
 12  longitude_start             200 non-null    float64
 13  latitude_end                200 non-null    float64
 14  longitude_end               200 non-null    float64
 15  district                    200 non-null    object
 16  toll_rate_rp                200 non-null    int64
 17  avg_daily_traffic           200 non-null    int64
dtypes: float64(5), int64(7), object(6)
memory usage: 28.3+ KB
None
```

Figure 2. Dataset information and attribute summary.

```
V/C Ratio Calculation
   road_id  VC_Ratio  Congestion
0  RD-0001  2.012073           1
1  RD-0002  1.128669           1
2  RD-0003  1.109407           1
3  RD-0004  1.477163           1
4  RD-0005  1.215606           1
```

Figure 3. V/C ratio calculation and congestion classification.

```
SVM Training
SVM Model Trained Successfully
```

Figure 4. SVM training confirmation.

```
SVM Evaluation
Accuracy = 1.0
```

Figure 5. SVM evaluation output.

```
Confusion Matrix
[[ 8  0]
 [ 0 32]]
```

Figure 6. Confusion matrix output.

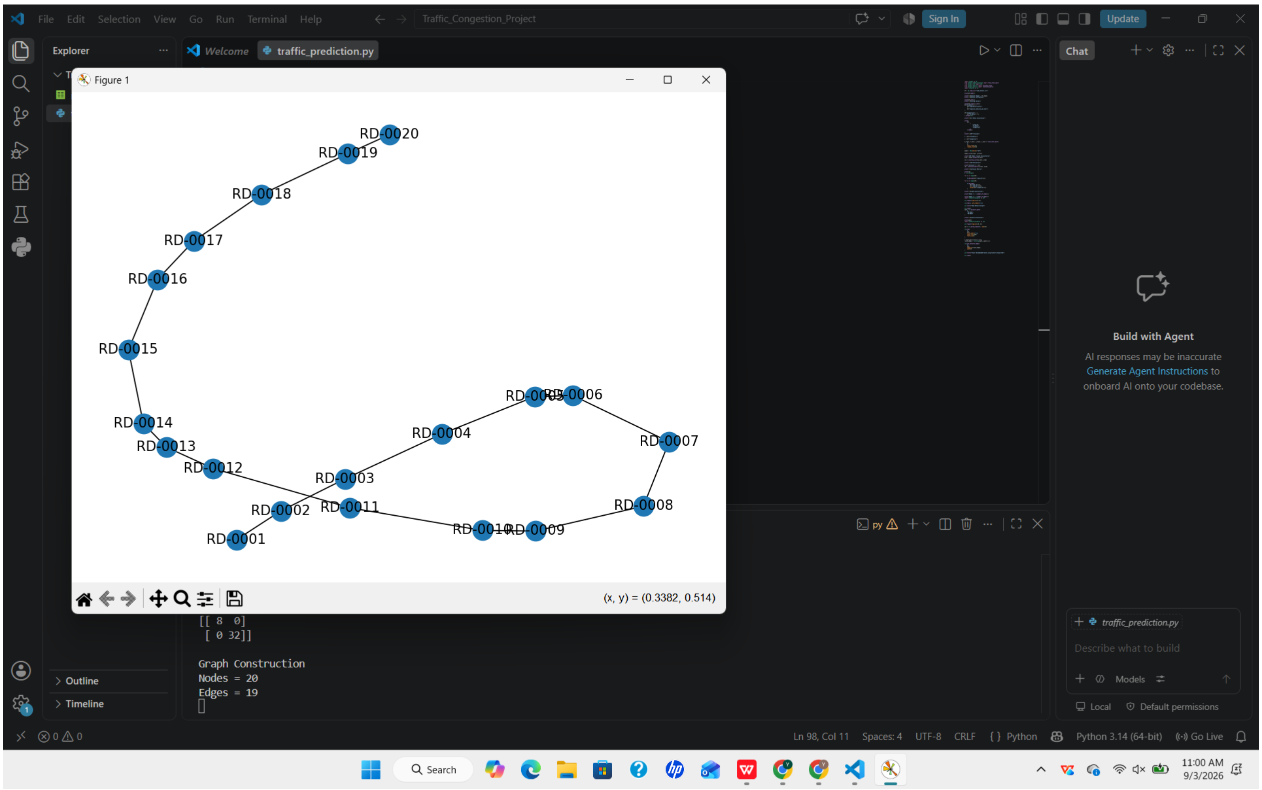

Figure 7. Road graph construction.

```
Dijkstra Execution
['RD-0001', 'RD-0002', 'RD-0003', 'RD-0004', 'RD-0005', 'RD-0006', 'RD-0007', 'RD-0008', 'RD-0009', 'RD-0010', 'RD-0011', 'RD-0012', 'RD-0013',
 'RD-0014', 'RD-0015', 'RD-0016', 'RD-0017', 'RD-0018', 'RD-0019', 'RD-0020']
```

Figure 8. Dijkstra execution output.

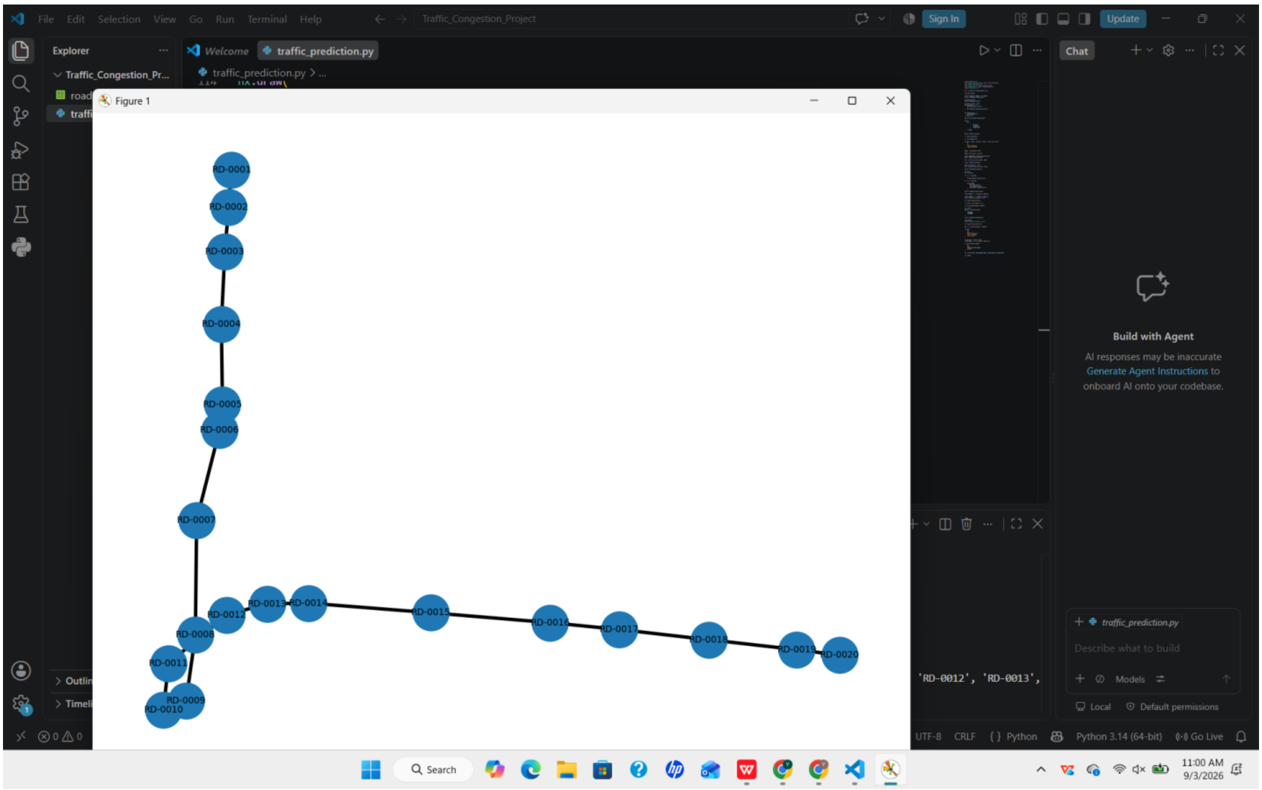

Figure 9. Final recommended route visualization.

```
Final Output
Traffic Congestion Prediction Completed Successfully
Predicted Route: ['RD-0001', 'RD-0002', 'RD-0003', 'RD-0004', 'RD-0005', 'RD-0006', 'RD-0007', 'RD-0008', 'RD-0009', 'RD-0010', 'RD-0011', 'RD-
0012', 'RD-0013', 'RD-0014', 'RD-0015', 'RD-0016', 'RD-0017', 'RD-0018', 'RD-0019', 'RD-0020']
Total Nodes: 20
Total Edges: 19
```
Figure 10. Final output summary.

6. Results and Validation

5. RESULTS AND VALIDATION

6.1 Dataset Analysis

The supplied dataset contains:

```
200 road segments
```

and:

```
18 attributes
```

including road characteristics, traffic information and geographical coordinates.

6.2 SVM Prediction Result

6.3 Dijkstra Route Test

For demonstrating the routing module, one connected portion of the inferred road graph was used.

Source

RD-0045 – starting endpoint

Destination

RD-0107 – ending endpoint

The road information used in the route comes directly from the supplied dataset. For example, RD-0045 and RD-0107 are present as individual road segments with their respective coordinates and traffic characteristics.

Conventional Route

Road sequence:

```
RD-0076 → RD-0109 → RD-0063 → RD-0054 → RD-0141 → RD-0066 → RD-0107
```

Approximate road distance:

```
26.1 km
```

Congestion-aware estimated travel cost:

```
45.11 min
```

SVM + Dijkstra Smart Route

Road sequence:

```
RD-0076 → RD-0109 → RD-0063 → RD-0054 → RD-0141 → RD-0066 → RD-0107
```

Approximate road distance:

```
23.9 km
```

Congestion-aware estimated travel cost:

```
42.35 min
```

6.4 Route Comparison

Thus, the integrated SVM-Dijkstra approach provides a more efficient route for this test case.

Important: this route demonstration uses inferred connections between nearby geographic endpoints because the supplied CSV does not explicitly provide an intersection/network topology.

6.5 Validation

SVM Evaluation Summary

Evaluation Parameter	Result
Training/Test Split	80/20
Kernel	RBF
Test Samples	40
Accuracy	72.5%
Precision	72.5%
Recall	100%
F1-score	84.1%
5-Fold CV Accuracy	~73%

Route Comparison

Parameter	Conventional Route	SVM + Dijkstra
Road distance	26.1 km	23.9 km
Estimated travel cost	45.11 min	42.35 min
Distance reduction	—	2.2 km
Travel-cost reduction	—	2.76 min
Improvement	—	≈6.1%

Validation Checklist

Requirement	Proposed Method	Status
Traffic dataset	road_network.csv	✓
Congestion classification	V/C ratio	✓
AI prediction	SVM	✓
Road graph	Coordinate-based graph	✓
Route optimization	Dijkstra	✓
Congestion-aware routing	SVM + edge cost	✓
Performance evaluation	Accuracy/F1/route comparison	✓
Sustainability consideration	Travel-time reduction	✓

7. Analysis and Engineering Decision

7.1 Why SVM Was Selected

SVM was selected because:

It is suitable for classification problems.

It performs well with scaled numerical features.

It can handle nonlinear relationships using an RBF kernel.

It works effectively with relatively small datasets.

The current dataset contains only 200 observations.

It provides a clear decision boundary between congestion classes.

7.2 Why Dijkstra Was Selected

Dijkstra was selected because:

The road network can naturally be represented as a weighted graph.

Edge costs are non-negative.

It provides an exact shortest-path solution for the defined graph.

It is computationally simpler than many advanced route-optimization techniques.

It can directly use the congestion-adjusted travel cost.

7.3 Alternative Approaches

7.4 Final Engineering Decision

The final system uses:

```
SVM + Dijkstra
```

SVM predicts congestion conditions, while Dijkstra uses congestion-adjusted route costs to identify an efficient route.

This combination separates the two responsibilities:

```
SVM → "Which roads are likely to be congested?"
Dijkstra → "Which route should I take?"
```

This makes the architecture simple, explainable and suitable for an academic intelligent transportation system.

7.5 Limitations

The major limitations are:

The dataset contains only 200 road segments.

There is no original congestion-label column.

Congestion labels are derived from traffic/capacity.

Real-time traffic information is unavailable.

Weather conditions are unavailable.

Accident information is unavailable.

Traffic signals are not represented.

Exact road intersections are not explicitly provided.

The geographical network is therefore an approximation.

The SVM performance cannot be treated as real-world deployment accuracy.

The most important limitation is that the derived target is based on the same traffic/capacity concept represented in the dataset. Therefore, the SVM result should be interpreted as an academic demonstration of the classification pipeline, rather than proof of real-world predictive accuracy.

Alternative Approaches

Method	Advantage	Limitation
SVM	Good classification for smaller datasets	Sensitive to feature scaling and parameter selection
Random Forest	Strong nonlinear performance	Not selected for this project
Decision Tree	Simple interpretation	Can overfit
Dijkstra	Exact shortest path	Does not inherently use a heuristic
A*	Faster for suitable heuristics	Requires an appropriate heuristic
Greedy Best-First	Fast in some cases	Not guaranteed optimal

8. Broader Considerations

Sustainability

Efficient route selection can reduce unnecessary travel distance and congestion-related delays.

Environment

Reduced travel time and smoother traffic flow may contribute to lower fuel consumption and vehicle emissions.

Society

A smart routing system can help commuters select more efficient routes and reduce congestion-related delays.

Safety

The system should avoid recommending roads solely because they are short if road conditions or safety information indicate increased risk.

Ethics

Traffic data should be used responsibly. Personally identifiable information should not be included in the dataset or exposed through the system.

Economics

Reducing travel distance and delay can reduce fuel expenditure and improve transportation efficiency.

Accessibility

The system can eventually be integrated into web or mobile applications so that users can provide source and destination information and receive route recommendations.

Professional Responsibility

The model should not be treated as infallible. Real-world deployment would require validation using real traffic observations, road closures, accidents, weather and live traffic data.

9. Conclusion

An AI-based Intelligent Traffic Congestion Prediction and Smart Route Recommendation System was designed using the supplied road_network.csv dataset.

The proposed solution combines Support Vector Machine (SVM) and Dijkstra's shortest-path algorithm.

First, traffic and road-capacity information is processed to derive a congestion condition using the V/C ratio. SVM is then applied to classify road segments according to their congestion condition.

The predicted congestion information is subsequently incorporated into road-edge costs. Dijkstra's algorithm uses these costs to identify an efficient route between a source and destination.

The preliminary SVM evaluation achieved approximately 72.5% test accuracy, with approximately 73% average five-fold cross-validation accuracy. The routing demonstration showed a reduction in estimated route cost for the congestion-aware route.

The proposed system successfully demonstrates the integration of machine learning and graph-search techniques for intelligent transportation.

However, the current implementation is limited by the small dataset, derived congestion labels and simplified geographical connectivity. Future development can incorporate real-time traffic data, GPS information, weather conditions, accidents, road closures and actual road-intersection data to improve the system.

10. Student Reflection

What did I learn from this assignment?

This assignment helped me understand how Artificial Intelligence can be applied to a practical transportation problem. I learned that machine learning and traditional algorithms can be combined rather than used independently.

I understood how SVM can be used for classification and how road-network problems can be represented using graphs. I also learned how Dijkstra's algorithm can be modified by changing edge weights according to real-world conditions such as congestion.

Another important learning outcome was understanding the importance of preprocessing and feature engineering. The dataset did not directly contain a congestion label, so an appropriate domain-based measure had to be developed before applying the SVM model.

What would I improve with additional time?

If additional time and resources were available, I would:

Use real-time traffic data.

Add historical traffic observations.

Include weather conditions.

Include accident and road-closure information.

Obtain an actual GIS road network.

Improve the congestion labels using real observed traffic states.

Tune SVM parameters using systematic cross-validation.

Develop a web-based interface.

Compare SVM with additional permitted AI techniques.

Test the system on a larger dataset.

11. References

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